

SQL Scenario-Based Assessment (DDL, DML, DQL, DCL & TCL)

Scenario 1 - Employee Management System

Scenario

You have joined an IT company as a Database Developer.

The HR team wants to maintain employee information.

Create an **Employee** table and insert the given records.

Later, HR asks you to display all employee details.

Table to Create

Employee

Column	Data Type
EmpID	INT
EmpName	VARCHAR(50)
Department	VARCHAR(30)
Salary	INT

Records to Insert

EmpID	EmpName	Department	Salary
101	John	IT	45000
102	Mary	HR	35000
103	David	Finance	55000
104	Sam	IT	40000

```
create database assessment;
```

```
/*SCENARIO 01*/  
create table assessment.employee(  
empid integer primary key,  
empname varchar(50),  
department varchar(30),  
salary integer);
```

```
select * from assessment.employee;
```

```
insert into assessment.employee(empid,  
empname,department,salary)  
values (101,'john','IT',45000),  
(102,'mary','HR',35000),  
(103,'david','finance',55000),  
(104,'sam','IT',40000),  
(105,'priya','HR',38000);
```

```
select * from assessment.employee;
```

EmpID	EmpName	Department	Salary
105	Priya	HR	38000

Question

Display all employee records.

SQL Command Type: DQL

Scenario 2 - College Student Database

Scenario

A college wants to maintain student details.

Create a **Student** table and insert the following records.

Later, the college decides to add an Email column.

Table

Student

Column	Data Type
--------	-----------

StudentID	INT
-----------	-----

StudentName	VARCHAR(50)
-------------	-------------

Course	VARCHAR(30)
--------	-------------

```
create table assessment.student(  
stu_id integer,  
stu_name varchar(50),  
course varchar(30));
```

```
select *from assessment.student;
```

```
insert into  
assessment.student(stu_id,stu_name,course)  
values (1,'Rahul','SQL'),  
(2,'Priya','Python'),  
(3,'Arun','Power BI'),  
(4,'Sneha','Java'),  
(5,'Karthik','SQL');
```

Records

StudentID	StudentName	Course
-----------	-------------	--------

1	Rahul	SQL
---	-------	-----

2	Priya	Python
---	-------	--------

3	Arun	Power BI
---	------	----------

```
select *from assessment.student;
```

```
alter table assessment.student  
add email varchar(50);
```

```
select *from assessment.student;
```

StudentID	StudentName	Course
-----------	-------------	--------

4	Sneha	Java
---	-------	------

5	Karthik	SQL
---	---------	-----

Question

Modify the Student table by adding a new column called **Email**.

SQL Command Type: DDL

Scenario 3 - Online Shopping System

Scenario

An online shopping company stores product information.

Create a **Product** table and insert the following records.

Later, the manager wants to increase the price of Product ID 103.

Table

Product

Column	Data Type
--------	-----------

ProductID	INT
-----------	-----

ProductName	VARCHAR(50)
-------------	-------------

Price	INT
-------	-----

Records

ProductID	ProductName	Price
-----------	-------------	-------

101	Laptop	60000
-----	--------	-------

102	Mouse	800
-----	-------	-----

```
/*SCENARIO 03*/
```

```
create table assessment.product(  
productID integer,  
productname varchar(50),  
price integer);
```

```
select * from assessment.product;
```

```
insert into assessment.product(productid,productname, price)  
values(101,'laptop',60000),  
(102,'mouse',800),  
(103,'keyboard',1200),  
(104,'monitor',15000),  
(105,'printer',9000);
```

```
select * from assessment.product;
```

```
update assessment.product  
set price=1500 where productid=103;
```

```
select * from assessment.product;
```

ProductID	ProductName	Price
103	Keyboard	1200
104	Monitor	15000
105	Printer	9000

Question

Update the price of Product ID 103 to **1500**.

SQL Command Type: DML

Scenario 4 - Hospital Database

Scenario

A hospital stores patient details.

Create a **Patient** table and insert the following records.

Later, Patient ID 2 requests that their record be removed from the system.

Table

Patient

Column	Data Type
PatientID	INT
PatientName	VARCHAR(50)
Disease	VARCHAR(50)

Records

PatientID	PatientName	Disease
1	Ramesh	Fever

```

/*SCENARIO 04*/
create table assessment.patient(
patientid integer,
patientname varchar(50),
disease varchar(50));

select * from assessment.patient;

insert into assessment.patient( patientid,patientname,disease)
values(1,'ramesh','fever'),
(2,'suresh','cold'),
(3,'Anitha','diabites'),
(4,'Meena','Asthama'),
(5,'kumar','typhoid');

select * from assessment.patient;

delete from assessment.patient
where patientid=2;

select * from assessment.patient;

```

PatientID	PatientName	Disease
-----------	-------------	---------

2	Suresh	Cold
3	Anitha	Diabetes
4	Meena	Asthma
5	Kumar	Typhoid

Question

Delete Patient ID 2 from the table.

SQL Command Type: DML

Scenario 5 - School Management System

Scenario

A school created a temporary table for annual event registration.

After the event, the school wants to remove all records but keep the table for next year.

Table

EventRegistration

Column	Data Type
StudentID	INT
StudentName	VARCHAR(50)
EventName	VARCHAR(50)

Records

StudentID	StudentName	EventName
1	Rahul	Dance

StudentID StudentName EventName

2	Priya	Singing
3	Arun	Drawing
4	Sneha	Quiz
5	Karthik	Drama

Question

Remove all records from the table without deleting the table structure.

SQL Command Type: DDL

```
/*SCENARIO 05*/
```

```
create table assessment.eventregistration(  
studentid integer,  
studentname varchar(50),  
eventname varchar(50));
```

```
select * from assessment.eventregistration;
```

```
insert into assessment.eventregistration(studentid,  
studentname, eventname)  
values (1,'rahul','dance'),  
(2, 'priya','singing'),  
(3,'arun','drawing'),  
(4,'sneha','quiz'),  
(5,'karthik','drama');  
select * from assessment.eventregistration;
```

```
truncate table assessment.eventregistration;
```

Scenario 6 - Company Database

Scenario

The company wants to rename its existing Employee table to Staff.

Table

Employee

EmpID EmpName Salary

101	John	45000
102	Mary	35000
103	David	50000
104	Sam	42000
105	Priya	39000

```
/*SCENARIO 06*/
```

```
create table assessment.employee1(  
empid integer,  
empname varchar(20),  
salary integer);
```

```
select * from assessment.employee1;
```

```
insert into assessment.employee1(empid,empname,salary)  
values(101,'john',45000),  
(102,'mary',35000),  
(103,'david',50000),  
(104,'sam',42000),  
(105,'priya',39000);  
select * from assessment.employee1;
```

```
rename table assessment.employee1 to assessment.staff;  
select * from assessment.staff;
```

Question

Rename the Employee table to **Staff**.

Scenario 7 - Banking System

Scenario

A bank manager updates a customer's account balance.

The manager wants the changes to be permanently saved in the database.

Table

Account

Column	Data Type	/*SCENARIO 07*/ create table assessment.account(accountno integer, customername varchar(50), balance integer); select * from assessment.account; insert into assessment.account(accountno,customername,balance) values (1001,'john',50000), (1002,'mary',30000), (1003,'david',70000), (1004,'sam',45000), (1005,'priya',60000); select * from assessment.account;
AccountNo	INT	
CustomerName	VARCHAR(50)	
Balance	INT	

Records

AccountNo	CustomerName	Balance	update assessment.account set balance=35000 where accountno=1002; commit; select * from assessment.account;
1001	John	50000	
1002	Mary	30000	
1003	David	70000	
1004	Sam	45000	
1005	Priya	60000	

Question

Update Account No. 1002's balance to **35000** and permanently save the changes.

Scenario 8 - Library Management System

Scenario

A librarian accidentally deletes one book record.

Before committing the transaction, they realize the mistake and want to restore the deleted record.

Table

Book

Column	Data Type	/*SCENARIO 08*/ create table assessment.book(bookid integer, bookname varchar(50), author varchar(50)); select * from assessment.book;
BookID	INT	
BookName	VARCHAR(50)	
Author	VARCHAR(50)	insert into assessment.book(bookid,bookname,author) values (1, 'sqlbasics','james'), (2,'python guide','robert'), (3,'java programming','john'), (4,'power BI','David'), (5,'data science','peter'); select * from assessment.book;

Records

BookID	BookName	Author	delete from assessment.book where bookid=3; rollback; /*unsuccessful*/
1	SQL Basics	James	
2	Python Guide	Robert	
3	Java Programming	John	
4	Power BI	David	
5	Data Science	Peter	

Question

Delete Book ID 3 and restore it before committing the transaction.

SQL Command Type: TCL

Scenario 9 - Company Security

Scenario

A new employee named **Student1** has joined the reporting team.

The manager wants the employee to view employee details but should not modify any records.

Existing Table

Employee

Sample Records

EmpID	EmpName	Department	Salary
-------	---------	------------	--------

101	John	IT	45000
-----	------	----	-------

102	Mary	HR	35000
-----	------	----	-------

103	David	Finance	50000
-----	-------	---------	-------

104	Sam	IT	42000
-----	-----	----	-------

105	Priya	HR	38000
-----	-------	----	-------

```
/*SCENARIO 09*/
```

```
create table assessment.employee2(  
empid integer,  
empname varchar(50),  
department varchar(50),  
salary integer);
```

```
select * from assessment.employee2;  
insert into assessment.employee2(empid,empname,department,salary)  
values(101,'john','IT',45000),  
(102,'mary','HR',35000),  
(103,'david','finance',50000),  
(104,'sam','IT',42000),  
(105,'Priya','HR',38000);
```

```
savepoint student1;
```

Question

Give **Student1** permission to view the Employee table only.

SQL Command Type: DCL

Scenario 10 - E-Commerce Database

Scenario

A junior developer accidentally creates a table called **CustomerDetails2026**.

The company decides to delete the entire table because it is no longer required.

Table

CustomerDetails2026

CustomerID	CustomerName	City
1	John	Chennai
2	Mary	Bangalore
3	David	Hyderabad
4	Sam	Coimbatore
5	Priya	Madurai

```
/*SCENARIO 10*/
```

```
create table assessment.customerdetails2026(  
customerid integer,  
customername varchar(50),  
city varchar(50));  
select * from assessment.customerdetails2026;
```

```
insert into assessment.customerdetails2026(customerid,  
customername,city)  
values(1,'john','chennai'),  
(2,'mary','banglore'),  
(3,'david','hydrabad'),  
(4,'sam','coimbatore'),  
(5,'priya','madurai');
```

```
drop table assessment.customerdetails2026;
```

Question

Delete the entire table permanently from the database.

SQL Command Type: DDL

Command Coverage

Scenario SQL Command Type

- | | |
|---|----------------|
| 1 | DQL (SELECT) |
| 2 | DDL (ALTER) |
| 3 | DML (UPDATE) |
| 4 | DML (DELETE) |
| 5 | DDL (TRUNCATE) |
| 6 | DDL (RENAME) |
| 7 | TCL (COMMIT) |
| 8 | TCL (ROLLBACK) |
| 9 | DCL (GRANT) |

Scenario SQL Command Type

10 DDL (DROP)